



FORT STREET HIGH SCHOOL

Name: _____

Teacher: _____

Class: _____

2016

HIGHER SCHOOL CERTIFICATE PRELIMINARY COURSE

ASSESSMENT TASK 3 – PRELIMINARY FINAL EXAM

Mathematics Extension 1

Time allowed: 2 Hours

(plus 5 minutes reading time)

Syllabus Outcomes	Assessment Area Description and Marking Guidelines	Questions
P3 PE3	chooses and applies appropriate arithmetic, algebraic, graphical , trigonometric and geometric techniques solves problems involving polynomials	1
P4	chooses and applies appropriate arithmetic, algebraic, graphical and trigonometric techniques	3, 4, 5
P6 P7	relates the derivative of a function to the slope of its graph determines the derivative of a function through routine application of the rules of differentiation	2
PE6	makes comprehensive use of mathematical language, diagrams and notation for communicating in a wide variety of situations	6

Total Marks 75

Attempt Questions 1-6

Question	Out of	Marks
Q1	/13	
Q2	/13	
Q3	/11	
Q4	/13	
Q5	/13	
Q6	/12	
Total	/75	
	Percent:	

General Instructions:

- Questions 1-6 are to be started in a new booklet
- The marks allocated for each question are indicated
- In Questions 1-6, show relevant mathematical reasoning and/or calculations.
- Marks may be deducted for careless or badly arranged work.
- Board – approved calculators may be used

Question 1 13 marks

marks

a. Find the exact value of $\tan 15^\circ$

2

b. Solve the inequality $\frac{x}{2x-1} \leq 5$

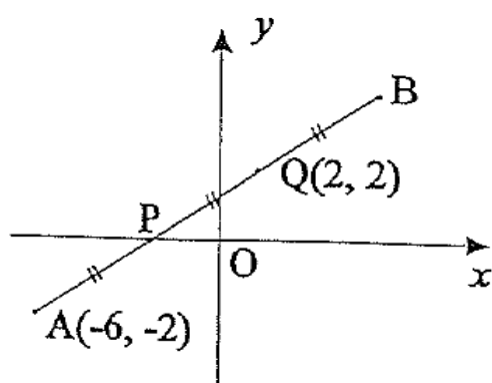
3

c. On the same axes, sketch the curve $y = x^2$ and $y = |x|$

2

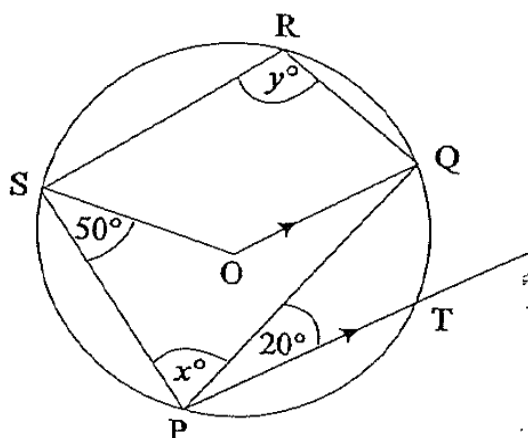
d. The diagram shows the line interval AB trisected at P and Q . The coordinates of A are $(-6, -2)$ and Q are $(2, 2)$. Find the coordinates of B .

2



e. In the diagram, O is the centre of the circle and $PT \parallel OQ$. Find the values of x and y , giving reasons.

4



a. Given that $f(x) = \frac{a^x + a^{-x}}{2}$, show that $f(2x) = 2[f(x)]^2 - 1$ 3

b. Differentiate $2 - 5x + 2x^2$ using first principles. 2

c. Differentiate each of the following 4

i. $\frac{1}{2}x^4 - 3x + 7$

ii. $\frac{2x}{x^2 + 4}$

iii. $x^2 \left(5x^3 - \frac{1}{2x^2} \right)^3$

d. Find the equation of the tangent to the curve $y = 2x\sqrt{x+1}$ at the point where $x = 3$. 4

Question 3 11 marks

marks

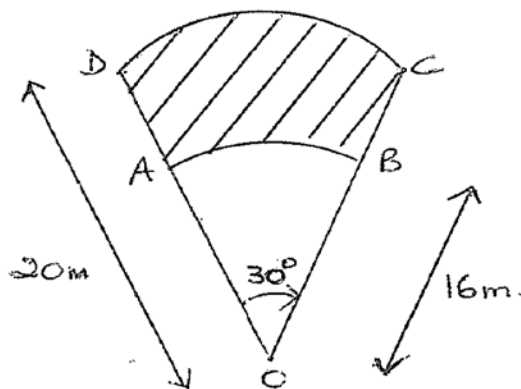
a.

If $\tan B = -0.2$ and $0^\circ < B < 270^\circ$, find the value of $\operatorname{cosec} B$ in surd form.

2

- b. The diagram below shows the field laid out for the shotput finals at Homebush. In order to qualify for the finals, the shotput must land in the shaded area $ABCD$.

4



Find the exact perimeter and area of the shaded area $ABCD$.

- c. Graph the curve $y = \sqrt{2} \sin x + \cos x$ for $0 \leq x \leq 2\pi$

3

- d. The lines $3x - y + 2 = 0$ and $mx - y - 1 = 0$ intersect at 45° . Find the possible value(s) of m .

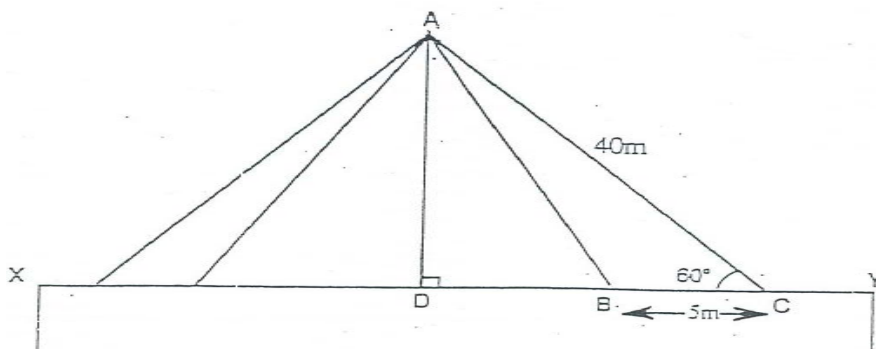
2

Question 4 13 marks

marks

- a. A horizontal bridge joins points X and Y . Cables were used to support the bridge as shown in the diagram. The distance between the cables AB and AC was $5m$ while AC was $40m$ and $\angle ACB = 60^\circ$

4



- i. Find the exact height of A above the bridge.
- ii. Use the cosine rule to find the length of the cable AB .

- b. Prove that $\frac{\sin 2x}{1 + \cos 2x} = \tan x$ and hence evaluate $\tan \frac{\pi}{8}$ in exact form.

3

- c. Find the general solutions for $\cos 2\theta = \cos \theta$

3

- d. Solve $\cos x - \sqrt{3} \sin x = 1$, where $0 \leq x \leq 2\pi$ by using the t method

3

Question 5 13 marks

marks

- a. 2
- i. Expand $\cos(\alpha + \beta)$

- ii. Show that $\cos 2\alpha = 1 - 2\sin^2 \alpha$

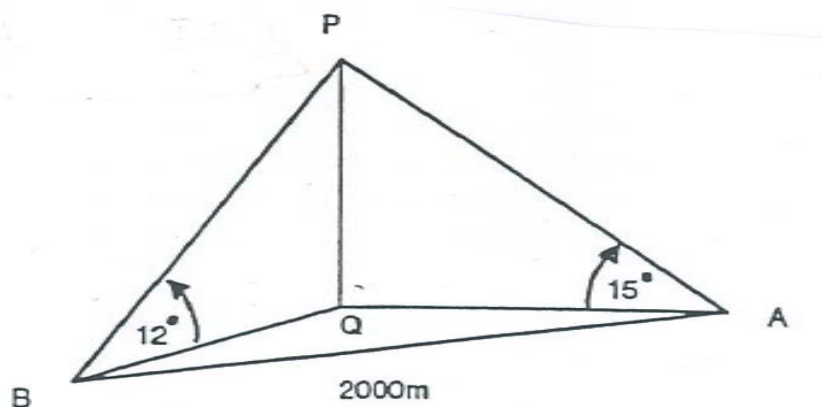
- b. If $\alpha = \tan^{-1}\left(\frac{5}{12}\right)$ and $\beta = \cos^{-1}\left(\frac{4}{5}\right)$, calculate the exact value of $\tan(\alpha - \beta)$ 2

- c. $ABCD$ is a cyclic quadrilateral.
Show that $\tan A + \tan B + \tan C + \tan D = 0$ 2

- d. 3
- i. Show that $\frac{\sin^3 \theta}{\cos \theta} + \sin \theta \cos \theta = \tan \theta$

- ii. Hence, or otherwise, solve $\frac{\sin^3 \theta}{\cos \theta} = 1 - \sin \theta \cos \theta$ for $0 \leq \theta \leq 360$

- e. 4



the angle of elevation of a tower PQ of height h metres at a point A due east of it is 15° . From another point B , due south of the tower the angle of elevation is 12° . The point A and B are 2000m apart on ground level. Find the height h of the tower to the nearest metre.

Question 6 12 marks

marks

- a. A man, M , is standing in a horizontal plane which also contains a tower, AB , which is 10 metres high, and a building, XY , which is 95 metres high. The man is 210 metres from the tower and his bearing from it is 42° . His bearing from the building is 110° . The bearing of the tower from the building is 150° . Put all this information into a diagram.

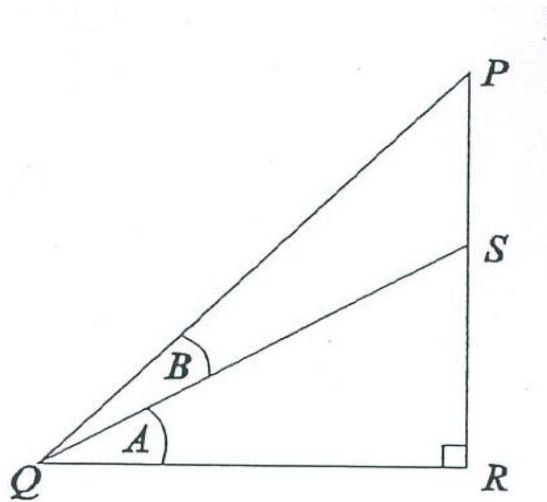
4

Find the angle of elevation from the top of the tower to the top of the building.
(to the nearest degree)

b.

In the diagram $\angle A > \angle B > 0$ and $\frac{\cos(A+B)}{\cos(A-B)} = \frac{4}{5}$

8



- i. Show that $\tan A \times \tan B = \frac{1}{9}$
- ii. If $PR = QR$, show that $9(\tan A + \tan B) = 8$
- iii. Hence, find A and B .



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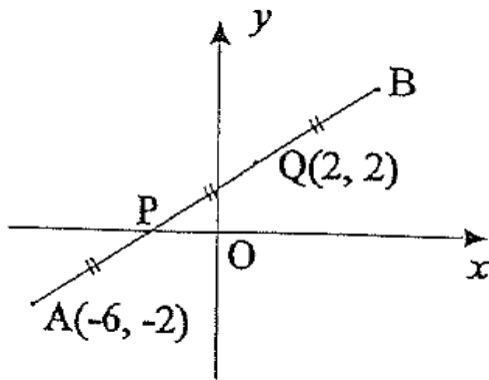
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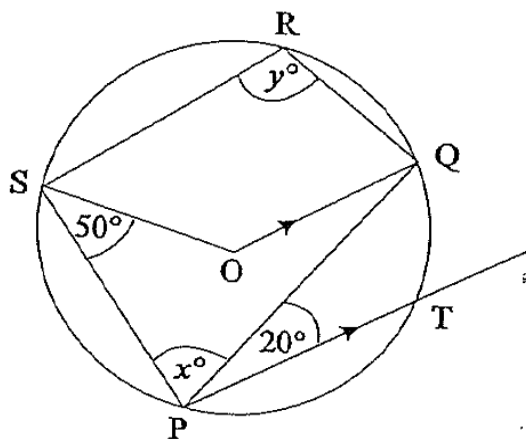
Question 1 13 marks	marks
a. Find the exact value of $\tan 15^\circ$ $\begin{aligned}\tan 15^\circ &= \tan(45 - 15) \\ &= \frac{\tan 45 - \tan 30}{1 + \tan 45 \tan 30} \\ &= \frac{\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{3}}}{1 + \frac{1}{\sqrt{2}} \frac{1}{\sqrt{3}}} \\ &= \frac{\sqrt{3} - \sqrt{2}}{\sqrt{6} + 1}\end{aligned}$ b. Solve the inequality $\frac{x}{2x-1} \leq 5$ $\begin{aligned}(2x-1)^2 \frac{2}{2x-1} &\leq 5(2x-1)^2 \\ (2x-1)x &\leq 5(2x-1)^2 \\ 0 &\leq 5(2x-1)^2 - x(2x-1) \\ 0 &\leq (2x-1)[5(2x-1) - x] \\ 0 &\leq (2x-1)(9x-5) \\ \frac{1}{2} > x &\geq \frac{5}{9} \quad \text{as } x \neq \frac{1}{2}\end{aligned}$ c. On the same axes, sketch the curve $y = x^2$ and $y = x$ $\begin{aligned}0 &< x < 1 \\ -1 &< x < 0\end{aligned}$ d. The diagram shows the line interval AB trisected at P and Q. The coordinates of A are $(-6, -2)$ and Q are $(2, 2)$. Find the coordinates of B.	2 3 2 2



Horizontal distance=4
vertical distance =2
Therefore $B(6, 4)$

- e. In the diagram, O is the centre of the circle and $PT \parallel OQ$. Find the values of x and y , giving reasons.

4



$x + y = 180$ (opposite angles of a cyclic quadrilateral sum to 180)
 $\angle QPT = \angle OQP = 20$ (alternate angles in parallel lines)
 $OP = OQ$ (radii of circle)
 $\triangle OPQ$ is isosceles (from above)
 $\therefore \angle OPQ = 20$ (base angles of isosceles triangles are equal)
 In $\triangle SOP$
 $SO = OP$ (radii of circle)
 $\triangle SOP$ is isosceles (from above)
 $\therefore \angle OPS = 50$ (base angles of isosceles triangles are equal)
 $\angle SPQ = \angle SPO + \angle QPO = 50 + 20 = 70$
 $x = 70, y = 180 - 70 = 110$

- a. Given that $f(x) = \frac{a^x + a^{-x}}{2}$, show that $f(2x) = 2[f(x)]^2 - 1$

3

$$\begin{aligned}
 f(2x) &= \frac{a^{2x} + a^{-2x}}{2} \\
 [f(x)]^2 &= \left[\frac{a^x + a^{-x}}{2} \right]^2 = \frac{a^{2x} + 2 \times a^x \times a^{-x} + a^{-2x}}{4} = \frac{a^{2x} + a^{-2x} + 2}{4} \\
 2[f(x)]^2 &= \frac{a^{2x} + a^{-2x}}{2} + 1 \\
 2[f(x)]^2 - 1 &= \frac{a^{2x} + a^{-2x}}{2} + 1 - 1 \\
 &= f(2x)
 \end{aligned}$$

- b. Differentiate $2 - 5x + 2x^2$ using first principles.

2

$$\begin{aligned}
 f(x) &= 2 - 5x + 2x^2 \\
 f(x+h) &= 2 - 5(x+h) + 2(x+h)^2 = 2 - 5x - 5h + 2x^2 + 4xh + 2h^2 \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{2 - 5x - 5h + 2x^2 + 4xh + 2h^2 - (2 - 5x + 2x^2)}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{-5h + 4xh + 2h^2}{h} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{h(-5 + 4x + 2h)}{h} \\
 f'(x) &= -5 + 4x
 \end{aligned}$$

- c. Differentiate each of the following

i. $\frac{1}{2}x^4 - 3x + 7$

4

$$2x^3 - 3$$

ii. $\frac{2x}{x^2 + 4}$

$$\frac{(x^2 + 4)2 - 2x(2x)}{(x^2 + 4)^2}$$

$$= \frac{-2x^2 + 8}{(x^2 + 4)^2}$$

iii. $x^2 \left(5x^3 - \frac{1}{2x^2} \right)^3$

$$x^2 \left(5x^3 - \frac{1}{2}x^{-2} \right)^3$$

$$= 2x \left(5x^3 - \frac{1}{2}x^{-2} \right)^3 + x^2 \left[3 \left(5x^3 - \frac{1}{2}x^{-2} \right)^2 (15x^2 + x^{-3}) \right]$$

$$= 2x \left(5x^3 - \frac{1}{2}x^{-2} \right)^3 + 3x^2 \left(5x^3 - \frac{1}{2}x^{-2} \right)^2 (15x^2 + x^{-3})$$

$$= x \left(5x^3 - \frac{1}{2}x^{-2} \right)^2 \left[2 \left(5x^3 - \frac{1}{2}x^{-2} \right) + 3x(15x^2 + x^{-3}) \right]$$

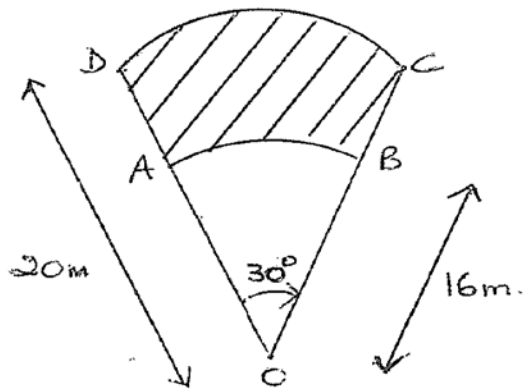
$$= x \left(5x^3 - \frac{1}{2}x^{-2} \right)^2 (10x^3 - 2x^{-2} + 45x^3 + 3x^{-2})$$

$$= x \left(5x^3 - \frac{1}{2}x^{-2} \right)^2 (55x^3 + x^{-2})$$

- d. Find the equation of the tangent to the curve $y = 2x\sqrt{x+1}$ at the point where $x = 3$.

$y = 2x(x+1)^{\frac{1}{2}}$ $y' = 2(x+1)^{\frac{1}{2}} + 2x \times \frac{1}{2}(x+1)^{-\frac{1}{2}} = 2\sqrt{x+1} + \frac{x}{\sqrt{x+1}}$ $x = 3: y' = 2\sqrt{4} + \frac{3}{\sqrt{4}}$ $\text{if } \sqrt{4} = 2 \quad y' = \frac{11}{2}$ $\text{if } \sqrt{4} = -2 \quad y' = -\frac{11}{2}$	
$x = 3, y = 2(3)\sqrt{4} = 12$ $y - 12 = \frac{11}{2}(x - 3)$ $11x - 2y - 9 = 0$	$x = 3, y = 2(3)\sqrt{4} = -12$ $y + 12 = -\frac{11}{2}(x - 3)$ $11x - 2y - 9 = 0$

Question 3	11 marks	marks
a. If $\tan B = -0.2$ and $0^\circ < B < 270^\circ$, find the value of $\operatorname{cosec} B$ in surd form.		2
Answer must be in the second quadrant $\sin B = \frac{1}{26}$ $\operatorname{cosec} B = \frac{1}{\sin B}$ $= \sqrt{26}$		4
b. The diagram below shows the field laid out for the shotput finals at Homebush. In order to qualify for the finals, the shotput must land in the shaded area $ABCD$.		



Find the exact perimeter and area of the shaded area $ABCD$.

$$l_1 = 16 \times \frac{\pi}{6} = \frac{8}{3}\pi$$

$$l_2 = 20 \times \frac{\pi}{6} = \frac{10}{3}\pi$$

$$P = \frac{8}{3}\pi + \frac{10}{3}\pi + 4 + 4 = 6\pi + 8$$

$$A = \frac{1}{2}(20)^2 \frac{\pi}{6} - \frac{1}{2}(16)^2 \frac{\pi}{6}$$

$$= \frac{1}{2} \frac{\pi}{6} (400 - 256)$$

$$= 144 \frac{\pi}{12} = 12\pi$$

3

2

c. Graph the curve $y = \sqrt{2} \sin x + \cos x$ for $0 \leq x \leq 2\pi$

d. The lines $3x - y + 2 = 0$ and $mx - y - 1 = 0$ intersect at 45° . Find the possible value(s) of m .

$$m_1 = 3, m_2 = m$$

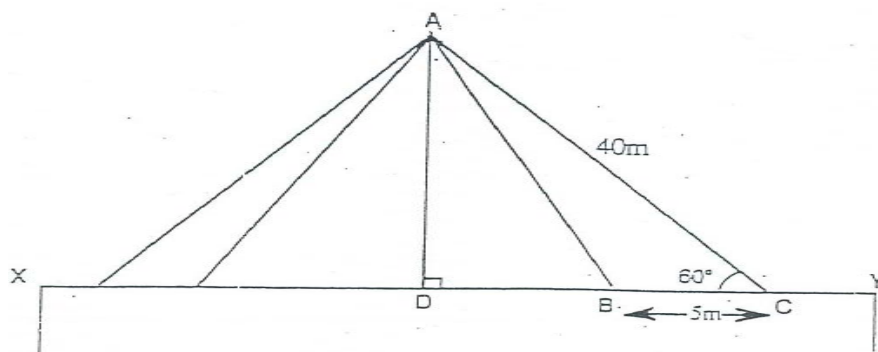
$$\tan \theta = \left| \frac{m-3}{1+3m} \right|$$

Question 4 13 marks

marks

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- i. Find the exact height of A above the bridge.
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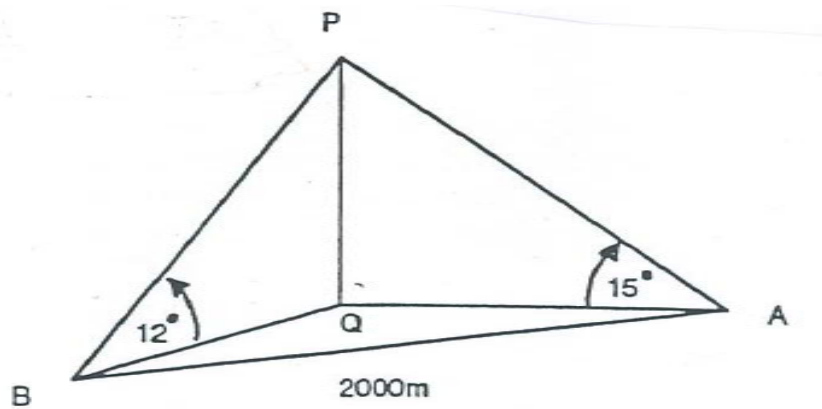
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- ii. Hence, or otherwise, solve $\frac{\sin^3 \theta}{\cos \theta} = 1 - \sin \theta \cos \theta$ for $0 \leq \theta \leq 360$

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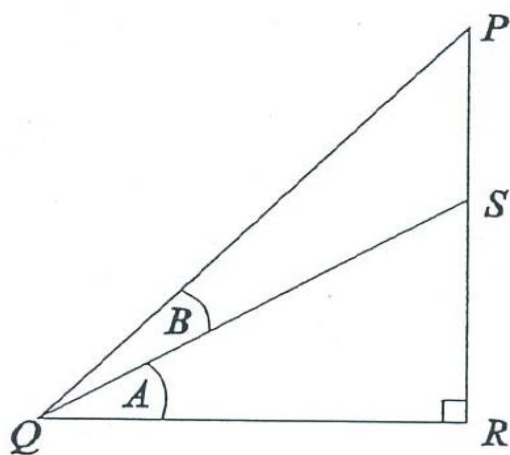
4

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b.

In the diagram $\angle A > \angle B > 0$ and $\frac{\cos(A+B)}{\cos(A-B)} = \frac{4}{5}$

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